Customer Shopping Behaviour Analysis

1. Project Overview:

The main goal of this project is to use transactional data to analyse consumer purchasing behaviour. The analysis's objectives are to find trends in consumer behaviour, comprehend consumer preferences, and offer information that can assist companies in enhancing their marketing plans and client interactions.

2.Dataset Summary:

The dataset contains data about consumer purchases and behaviour. Important features of the dataset include:

- Rows: 3,900
- Columns: 18
- Key Features:
    - Customer demographics (Age, Gender, Location, Subscription Status)
    - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Colour)
    - Shopping behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

This information offers a solid basis for analysing consumer purchasing patterns.

3.Exploratory Data Analysis using Python:

To ensure accuracy, the dataset was cleaned and prepared before analysis.

- **Data Loading** for data analysis and manipulation, the dataset was imported using Python (Pandas).
- **Initial Data Exploration** An initial investigation was carried out to comprehend Dataset arrangement, Data Types, Handling Missing Values Overview statistics, Missing values. Used df.info() to Check Structure and .describe() for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Handling Missing Values:** The *Review Rating* column contained 37 missing values. These values were filled using the median rating, which helps maintain the distribution of the data.
- **Column Standardization:** For improved readability and documentation, columns were renamed in **snake case**.
- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.
  - Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:** Promo_code_used was dropped after it was confirmed that discount_applied and promo_code_used were redundant.
- **Database Integration:** The cleaned Data Frame was loaded into the database for SQL analysis after a Python script was connected to MSSQL.

4. Data Analysis using SQL (Business Transactions):

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	revenue
1	Male	157890
2	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	purchase_amount
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91

Total rows: 839

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased	Average Product Rating
1	Gloves	3.860000
2	Sandals	3.840000
3	Boots	3.820000
4	Hat	3.800000
5	Skirt	3.780000

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

	shipping_type	(No column name)
1	Standard	58
2	Express	60

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

	subscription_status	total_customer	avg_spend	revenue
1	No	2847	59	170436
2	Yes	1053	59	62645

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
1	Hat	50.000000000000
2	Sneakers	49.660000000000
3	Coat	49.070000000000
4	Sweater	48.170000000000
5	Pants	47.370000000000

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number of Customers
1	Returning	701
2	Loyal	3116
3	New	83

8. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
1	1	Accessories	Jewelry	171
2	2	Accessories	Belt	161
3	3	Accessories	Sunglasses	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status	repeat_buyesr
1	Yes	958
2	No	2518

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group	total_revenue
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI:

Finally, we built an interactive dashboard in **Power BI** to present insights visually.

6. Business Recommendations:

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.
- **Customer Loyalty Programs** – Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.